

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS:

Total Marks: 30

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate **HTML5 semantic elements** missing in the structure.
2. Modify the structure to make it **standards-compliant**.
3. Interpret the **HTTP request generated** during form submission.
4. Analyze the **HTTP response cycle** from server to browser.
5. Suggest improvements for **usability and accessibility**.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
  <head>
    <title>Company</title>
  </head>
  <body>
    <div>
      <h1>Welcome
      <p>About us
    </div>
  </body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.
2. Analyze how different browsers may interpret this structure.
3. Identify **missing semantic elements** (e.g., header, section).
4. Provide a **corrected W3C-compliant HTML structure**.
5. Suggest techniques to ensure **cross-browser compatibility**.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

Questions:

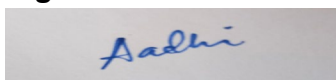
1. Identify issues in the **form configuration**.
2. Analyze why the **HTTP request is not triggered**.
3. Interpret the role of **GET method in this scenario**.
4. Propose a **corrected implementation**.
5. Suggest improvements for **secure data transmission**.

Code of Conduct:

I Aadhirai S P(192411002) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



HTML5 & HTTP Interactive Explorer

Explore modern web development concepts through beautiful visual representations of semantic HTML, HTTP communication, browser compatibility and secure form submission.

[Start Exploring !](#)

Cross-Browser Compatibility Issue

Original HTML

```
<html>
<head>
  <title>Company</title>
</head>
<body>
  <div>
    <h1>Welcome
```

Detected Problems

- ✗ Missing closing <h1>
- ✗ Missing closing <p>
- ✗ Missing semantic tags
- ✗ Poor accessibility
- ✗ Browser rendering differences

Correct HTML5 Structure

```
<header>
  <h1>Welcome</h1>
</header>
<main>
  <section>
```



Excellent HTML5 Support



Auto Corrects Invalid HTML



Chromium Rendering Engine



Requires Standards Compliance



Form Submission Failure

Problem

```
<form action="/submit"
method="GET">
<input type="text">
<input type="password">
<button
type="button">Login</button>
```

Issues Found

- ✗ Button is not submit
- ✗ GET exposes password
- ✗ HTTP request never triggered
- ✗ No validation
- ✗ Insecure transmission

Correct Solution

```
<form action="/submit"
method="POST">
<label>Username</label>
<input type="text"
required>

<label>Password</label>
<input type="password"
required>

<button
```



HTTP Status Dashboard

200 OK
Request Successful

201 Created
New Resource Created

404 Error
Resource Not Found

500 Error
Internal Server Error



HTTP Communication Timeline

1 User Opens Form
Browser loads HTML page.

2 User Enters Details
Name and Email are entered.

3 Validation



 **WebTech Explorer**

Portal Browser Forms

 **Best Practices**

Accessibility

- ARIA Labels
- Labels for Inputs
- Keyboard Navigation
- Screen Reader Support

Security

- HTTPS
- POST Method
- Password Encryption
- CSRF Protection

Compatibility

- HTML5 Validation
- Responsive CSS
- Cross Browser Testing
- Semantic HTML

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand